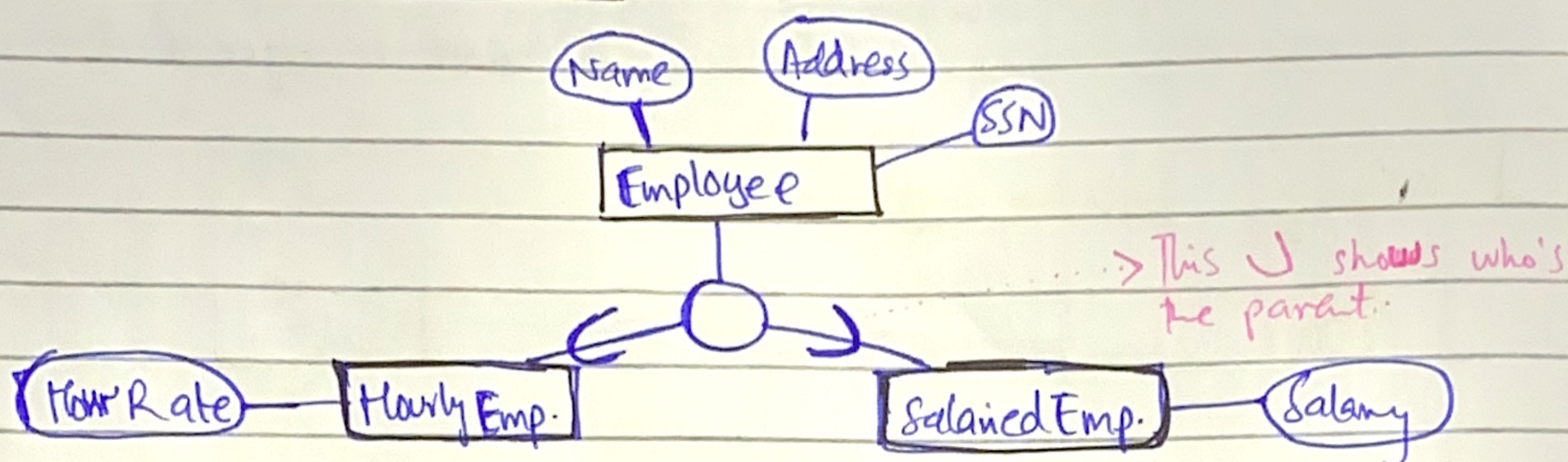


Slides: "8. EER modeling"

LECTURE - 21

Monday (online)
20/4/26.

Enhanced ER-model:



- "Employee" is Supertype (parent) with "Hourly Emp" and "Salaried Emp" as subtypes (childs). Showing inheritance where both childs have all the attributes of parent i.e. Employee AND must have at least one extra.

⇒ Total and Partial Constraint: Page # 18/44 & 19/44.

- Double line b/w parent & child shows that ^{instance} must be from either of the child of parent like: patient can either be "outpatient" or "resident".
- Single line shows that an instance of parent could be either of its child OR could be any other option too like: vehicle could be Car, truck or any other like motorbike.

⇒ Disjointness Constraints: Page # 21/44 & 22/44.

- 1) Disjoint: An instance could be ONLY one of the subtypes (cannot be both)
e.g: Patient can either be 'resident' or 'outpatient'
Represented by (d) in b/w the subtypes & supertype.

- 2) Overlap: An instance can be either one or all of the subtypes.
e.g: Employee could be both hourly & salaried at the same time. (for overtime)
Represented by (o)

Generalization: Converting multiple entities to have a new parent entity.
Specialization: Making child entities out of existing parent entity.

↳ problem statement: Page # 15-16 / 44

we've parts of vehicle that can be manufactured or purchased.

• On page 15/44 we made "supplier" as a multi-valued attribute that it can have either a supplier name (if parts are purchased) or NULL if parts are manufactured.

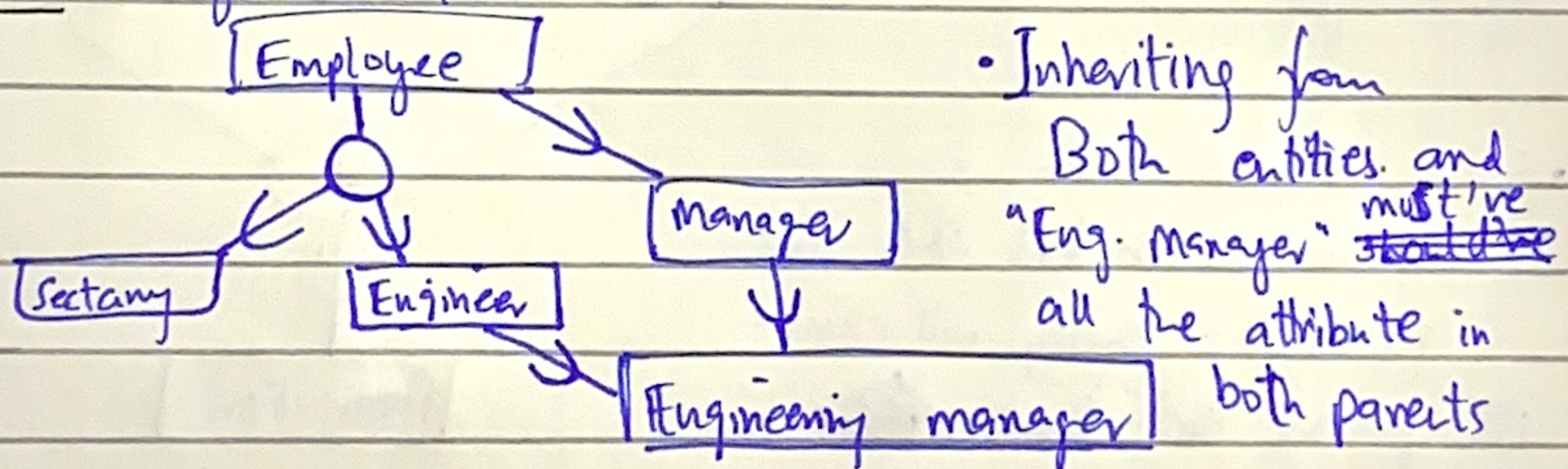
• That's wrong approach so we fixed in 16/44 by making a specialized subtype i.e: "purchased parts" & "manufactured parts".

So now only entity of "purchased parts" will have supplier entity as manufactured parts does not have suppliers.

NOTE: - Unit-Price is with Supplies and not supplier since unit price is of parts supplied; not of supplier itself.

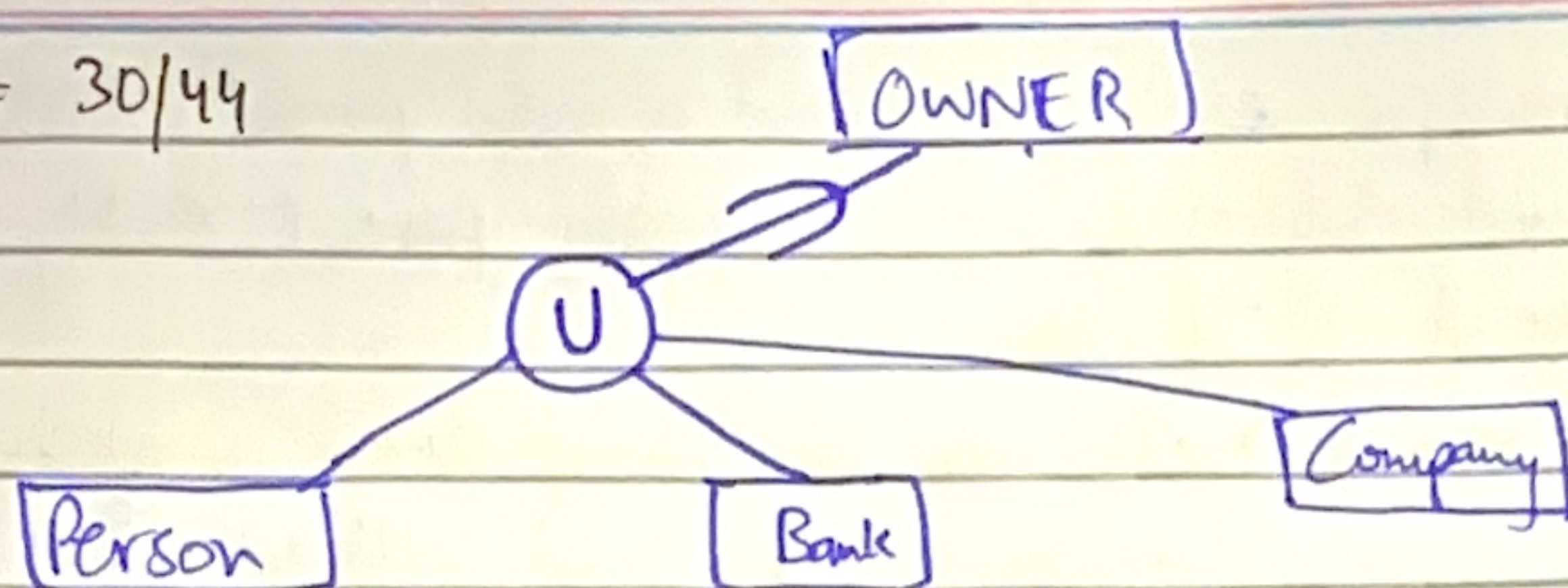
- Single line b/w PART and two child showing "partial constraint" as parts like: 30 wheels are manufactured & 20 wheels are purchased.

⇒ Shared Classes: Page # 26 / 44.



⇒ Full example diagram: Pg # 23 / 44.

⇒ UNION TYPES: Have multiple super types with single subtype. This shows that the subtype can choose 'any one' of super type as its parent at a time.



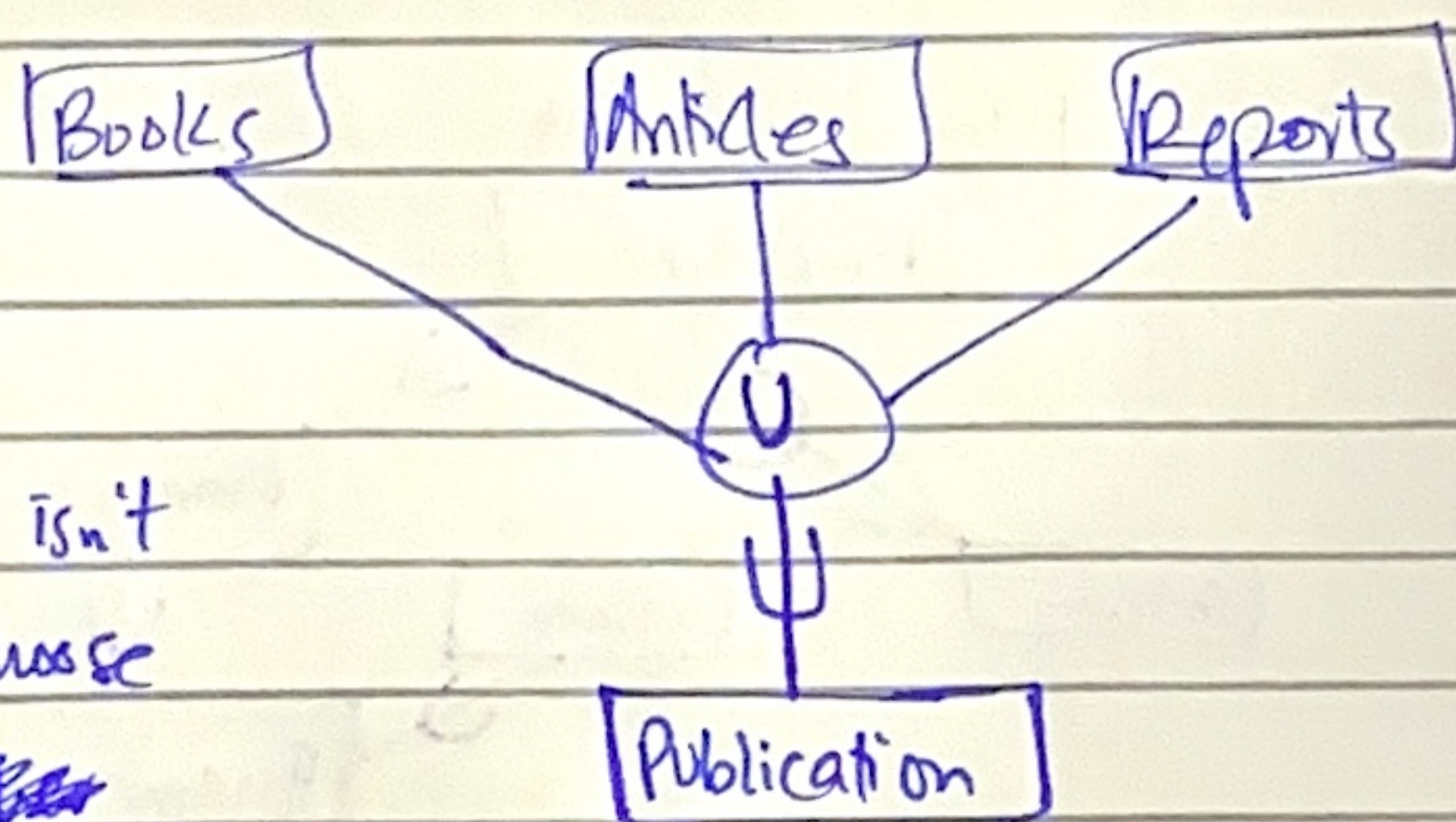
'Owner' is either the Bank, a person or a company.

Here the OWNER subclass is the union of 'Person', 'Bank' and 'Company' as its Parent/Super class means any one of these could be chosen.

→ Difference from "Shared classes": a share class has all of attributes of its superclasses where as here the sub-class have only one parent's attributes.

Q: We've list of Books, articles and Reports. Some of them are published while others are under process. Is it a Union type or Inheritance.

Union Type:



- Since some of written is ready for publish but other isn't ∴ publication entity will choose its parent itself that ~~whether~~ whether its instance is something ready for publish or not.

- If all were there for publish; then it would be Simple Inheritance with 'Publication' being Supertype & other three as subtypes.

⇒ Alternate Diagram Model Notations: Page 41-42/44.